

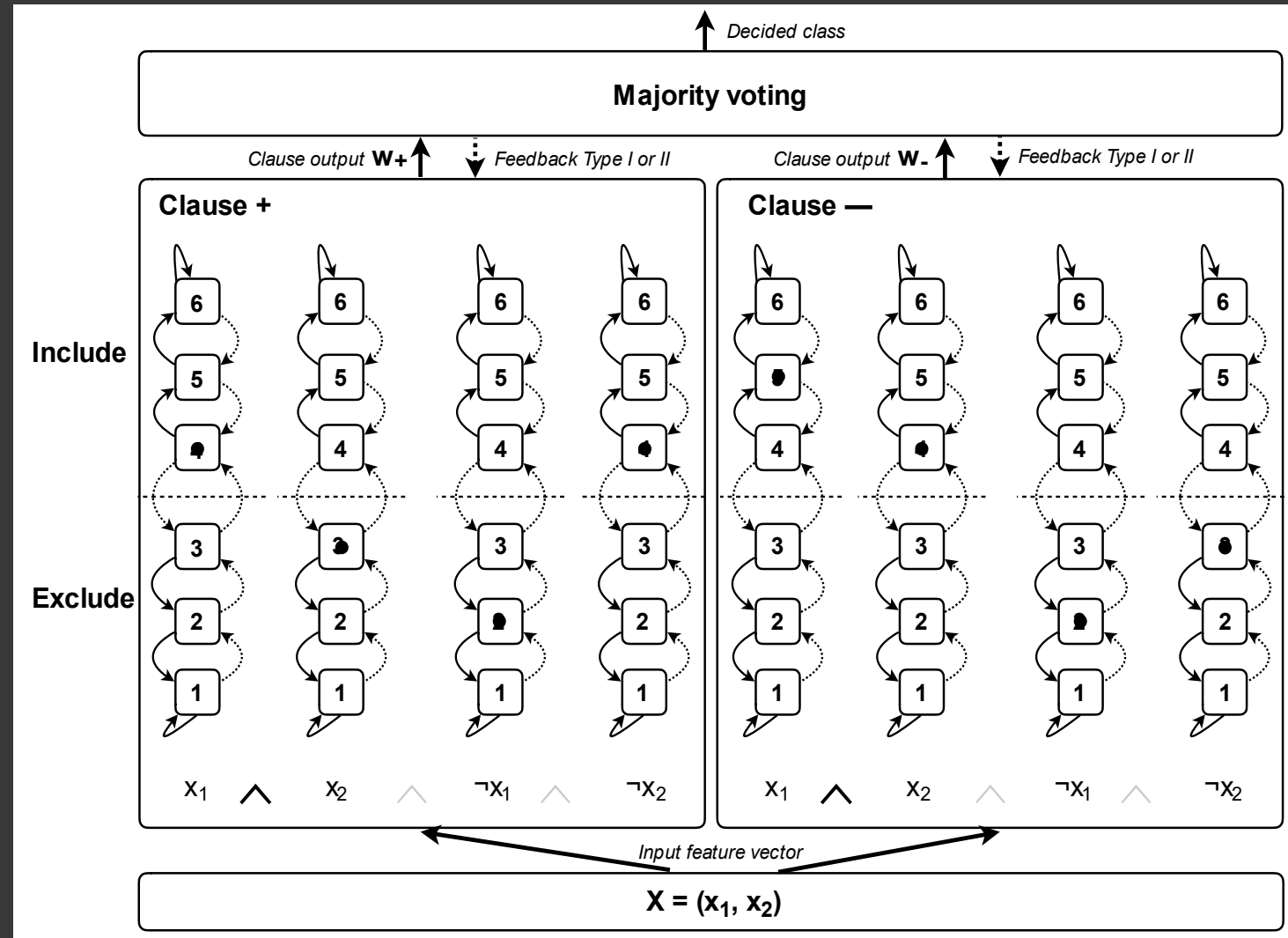
# Markov chain analysis of learning process

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# Overview

- Markov chains
- Single literal learning
- Literal games
- Multi-literal learning
- Multi-rule learning

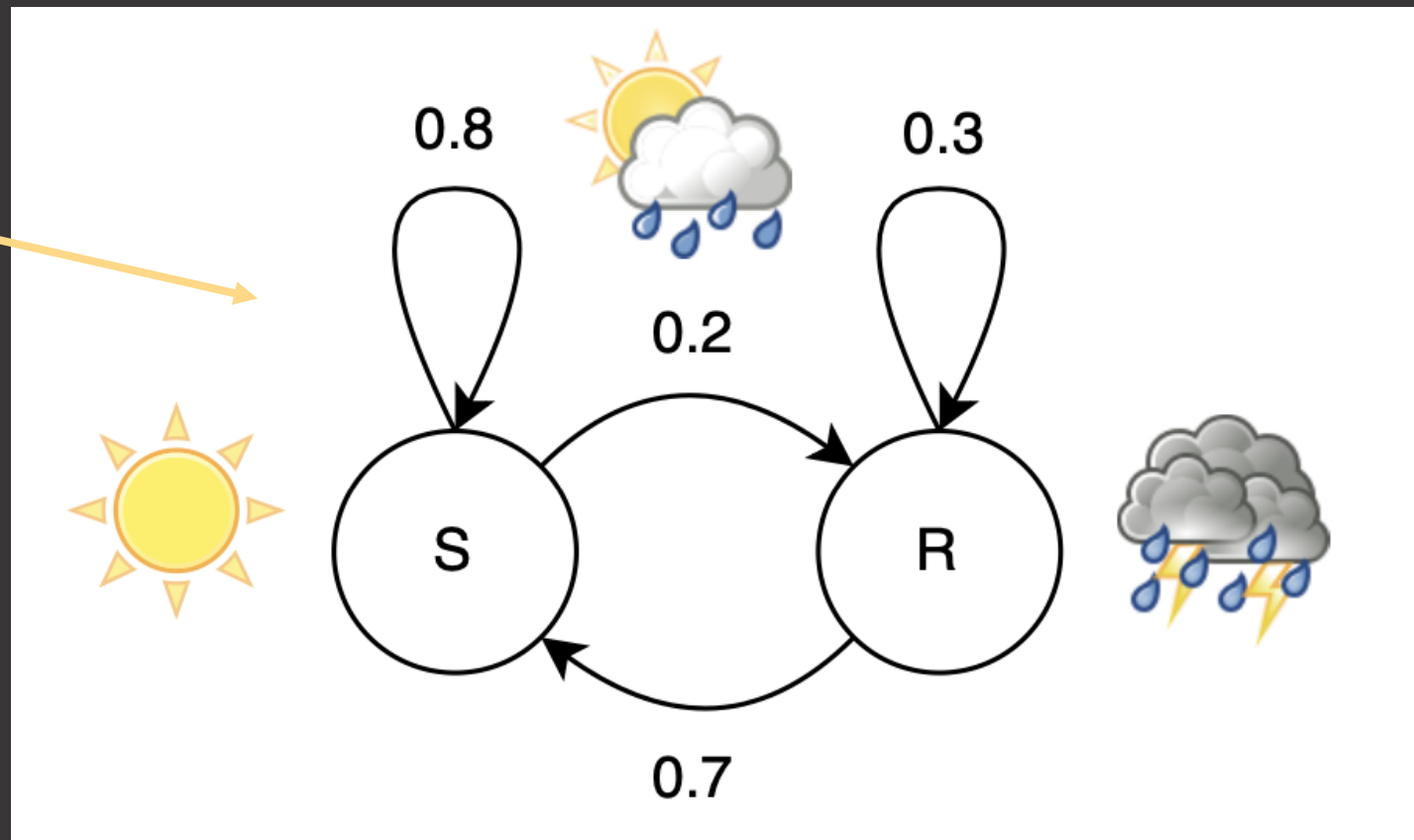
# Analysis of Tsetlin machine learning



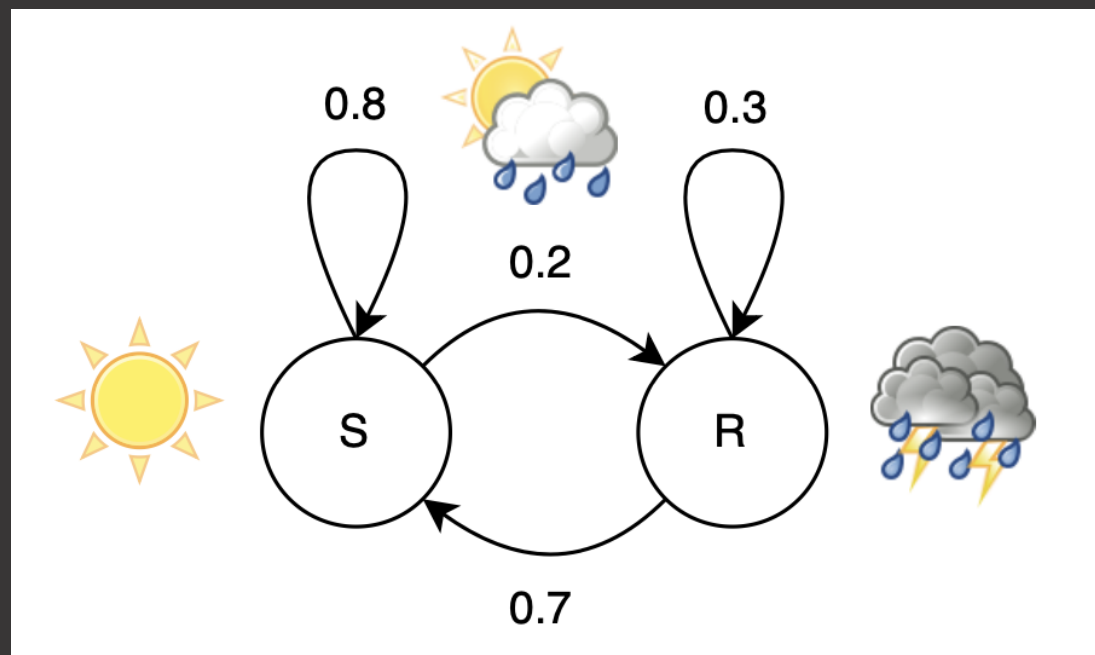
<https://github.com/cair/TsetlinMachine>

# Markov chain

*Transition graph*

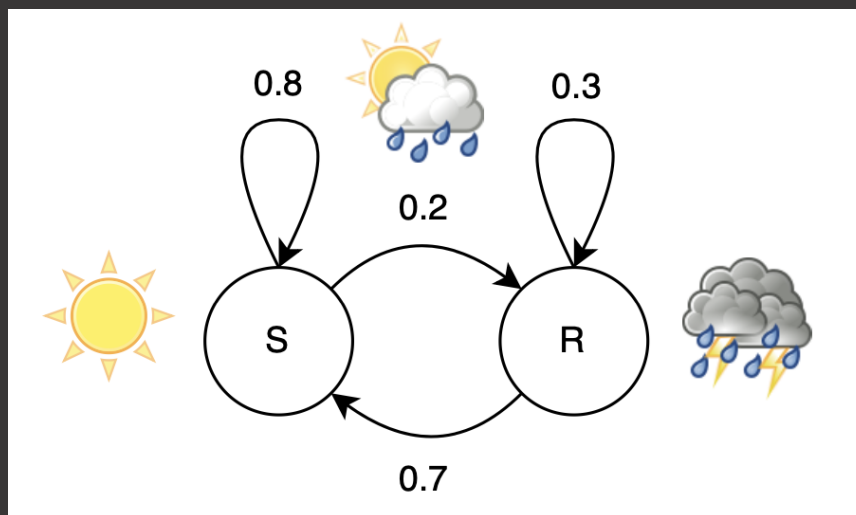


# Markov property



**Markov property:** The probability of a state transition only depends on the current state.

# Stationary distribution

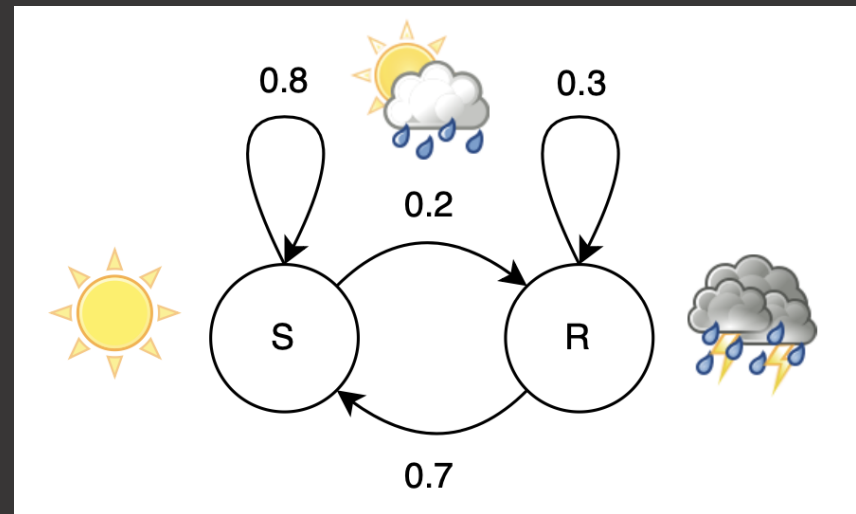


$$P(S) = 0.8 \cdot P(S) + 0.7 \cdot P(R)$$

$$P(R) = 0.2 \cdot P(S) + 0.3 \cdot P(R)$$

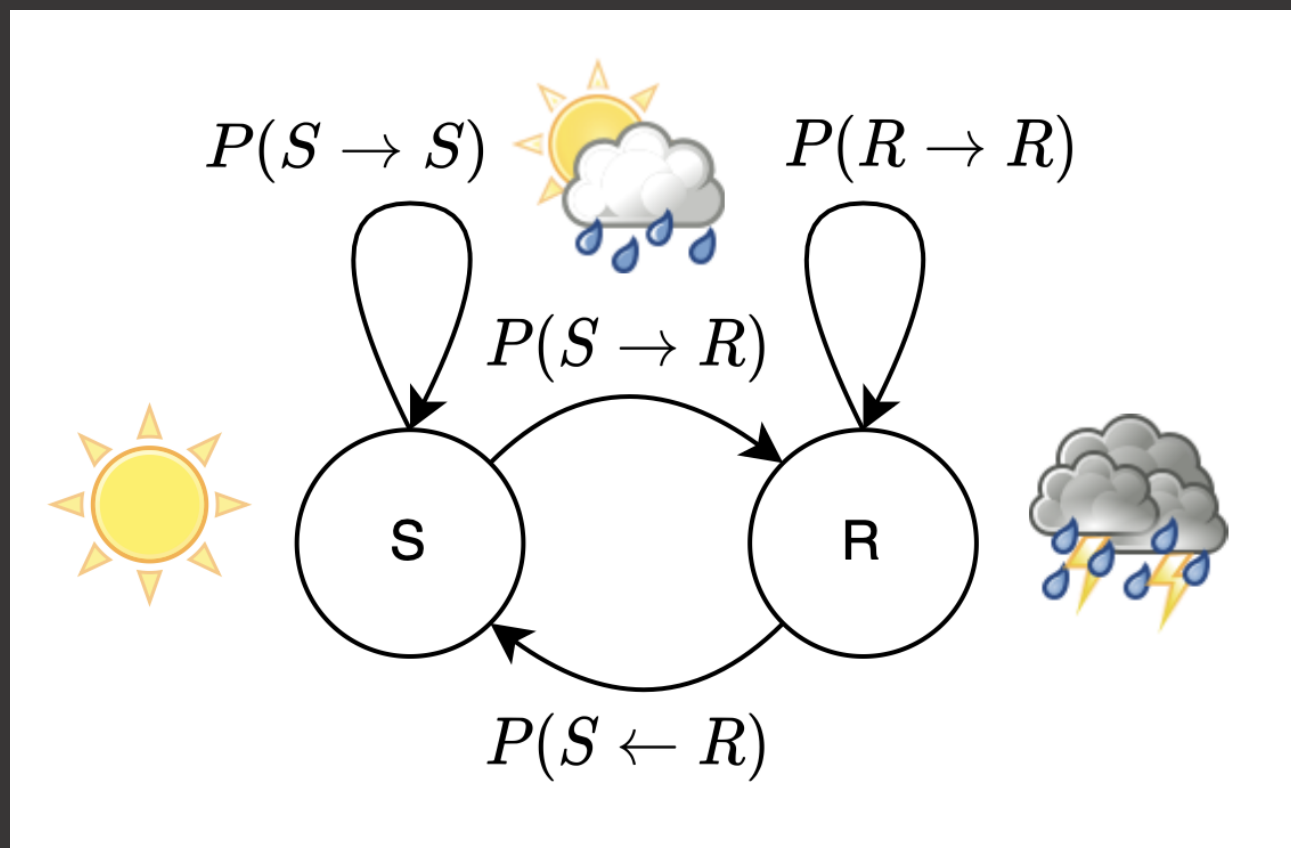
$$1 = P(S) + P(R).$$

# Stationary distribution



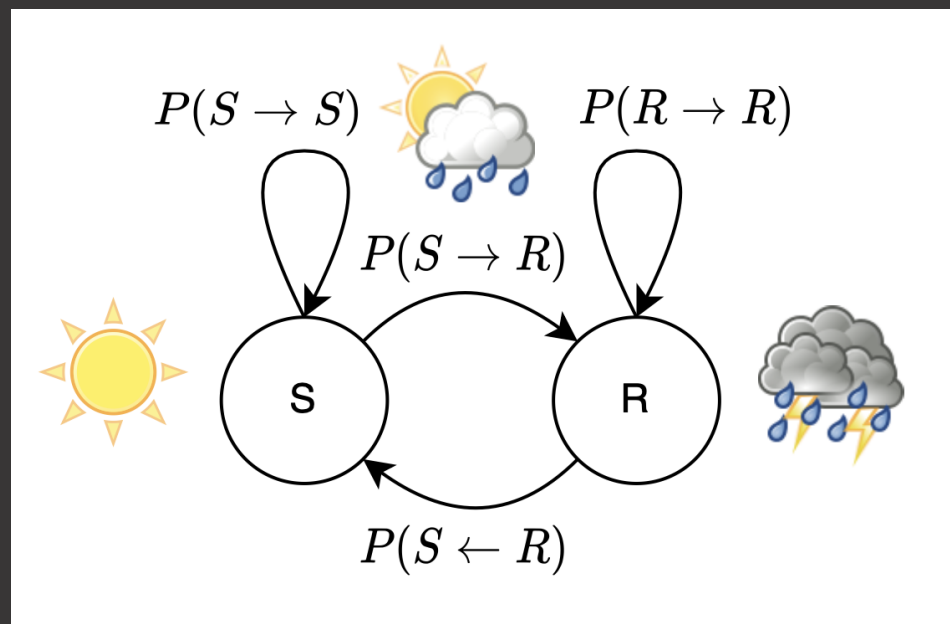
$$\begin{aligned}
 P(S) &\approx 0.78 \\
 P(R) &\approx 0.22.
 \end{aligned}$$

# Generic transition graph





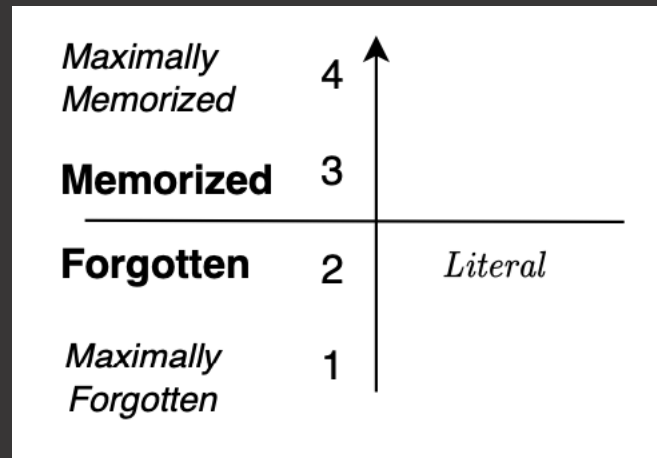
# Generic transition graph



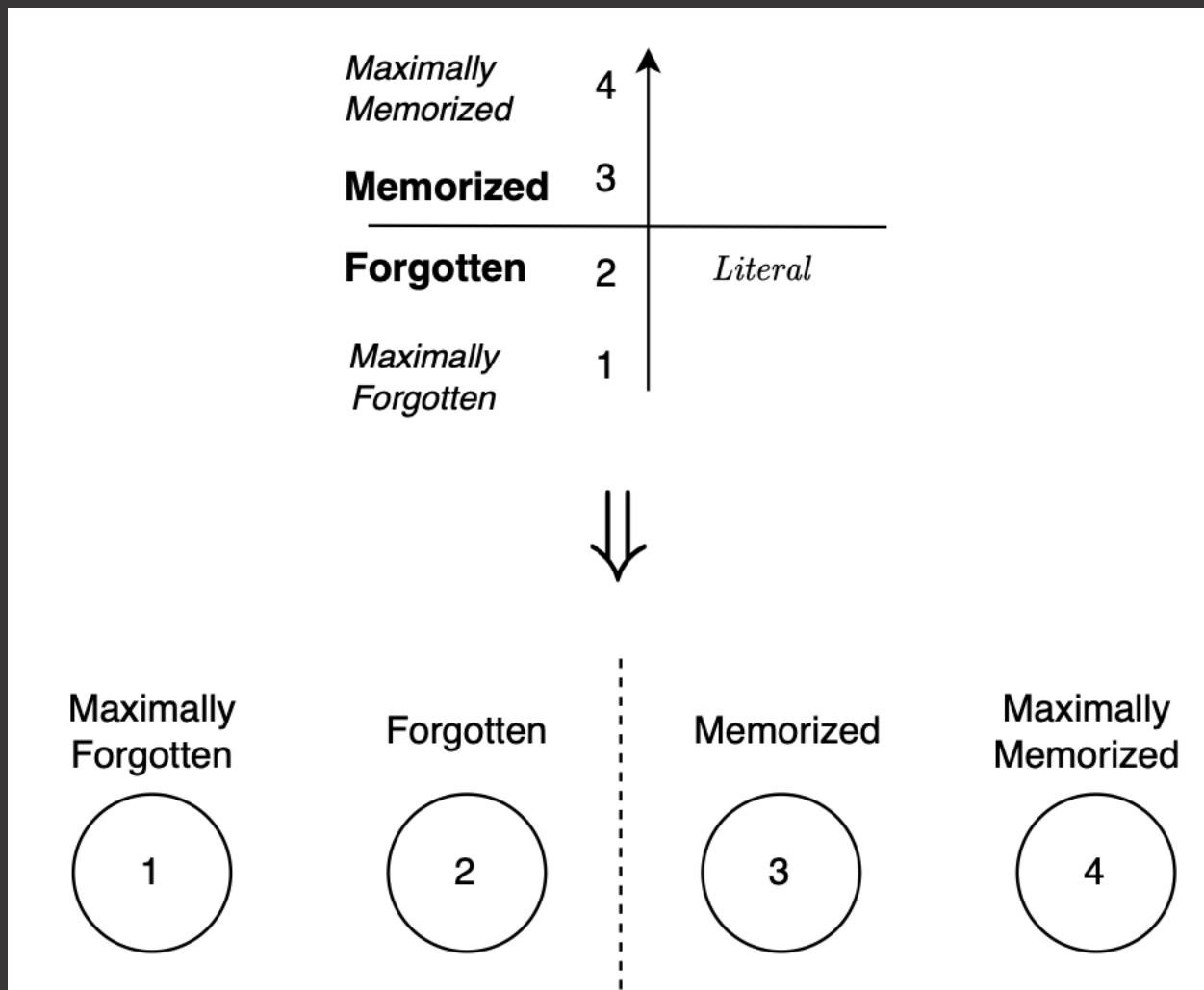
$$\begin{aligned}
 P(S) &= \frac{P(R \rightarrow S)}{P(R \rightarrow S) - P(S \rightarrow S) + 1} \\
 P(R) &= \frac{P(S \rightarrow R)}{P(R \rightarrow S) - P(S \rightarrow S) + 1} \cdot
 \end{aligned}$$

# Single literal learning

# Tsetlin machine memory

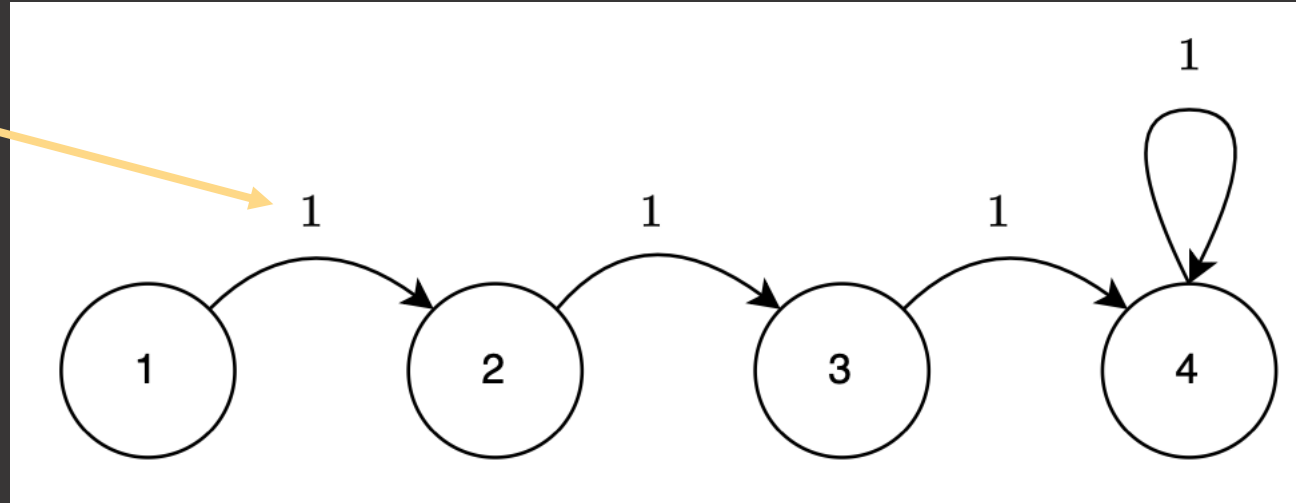


# Literal Automaton



# Target class: Type I Feedback (Recognize)

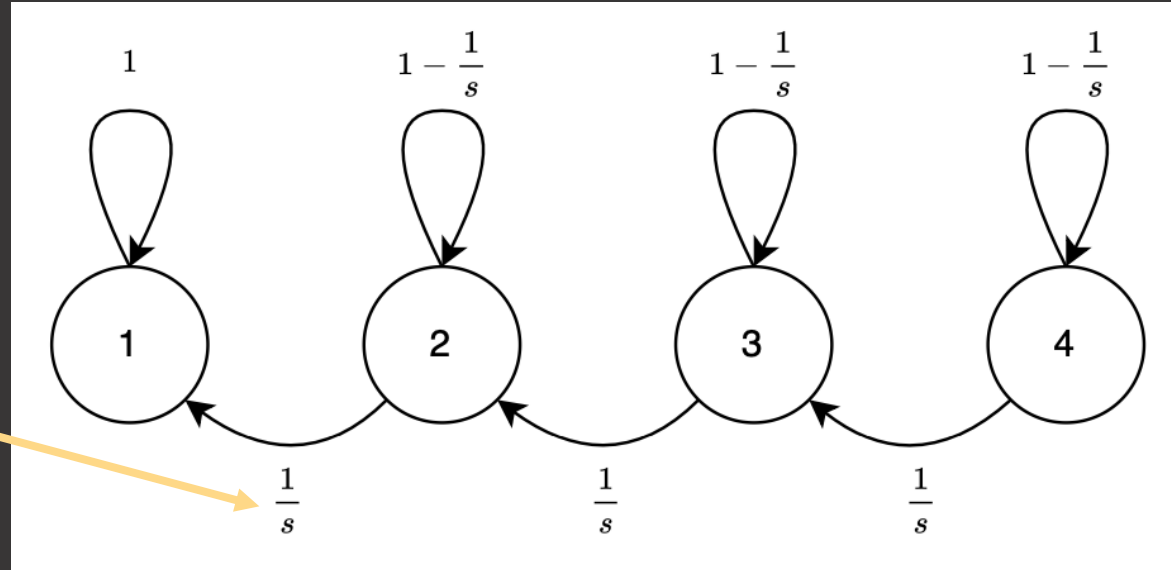
*Memorize Value*



- If the rule condition is *True*: memorize all *True* literals in the input
- That is, increment the literal's position in memory unless already *Maximally Memorized*

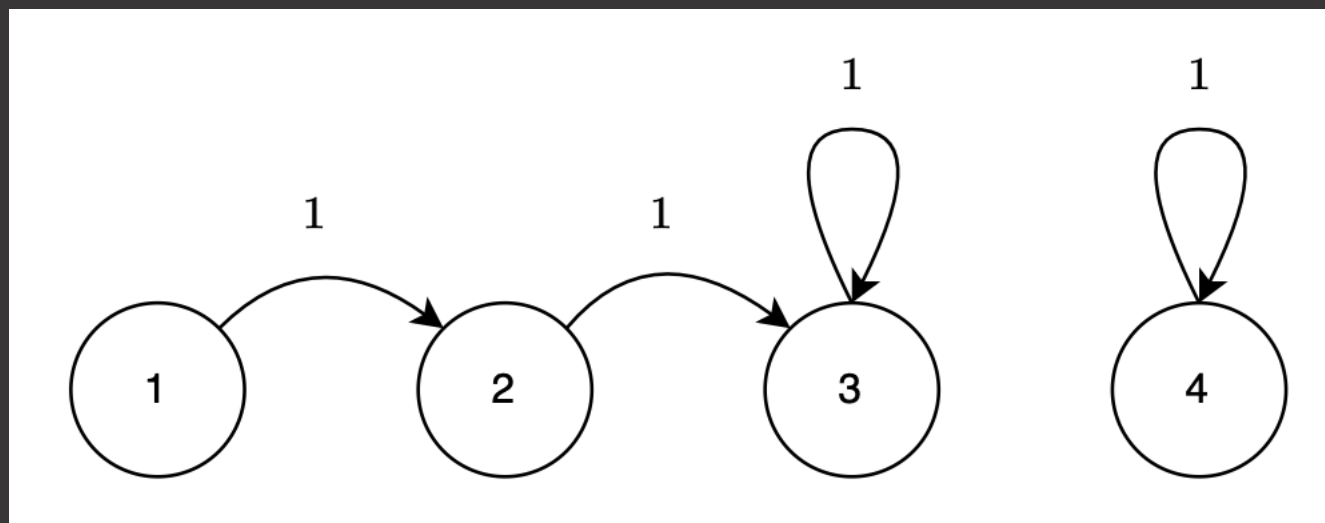
# Target Class: Type I Feedback (Erase)

Forget Value



- Forget all remaining literals by pushing them towards *Maximally Forgotten*.
- You forget a literal by decrementing its position with probability  $1/s$  unless *Maximally Forgotten*.

# Other Class: Type II Feedback (Reject)



If the rule condition is *True*: memorize all Forgotten literals that are *False*

# Observations drive learning

- Recognize* —→ 1. Observation  $(L, Y)$  means the literal is *True* and the class is  $Y$ .
- Erase* —→ 2. Observation  $(\bar{L}, Y)$  means the literal is *False* and the class is  $Y$ .
3. Observation  $(L, \bar{Y})$  means the literal is *True* and the class is  $\bar{Y}$ .
- Reject* —→ 4. Observation  $(\bar{L}, \bar{Y})$  means the literal is *False* and the class is  $\bar{Y}$ .



# Probability of observations

$$P(LY), P(\bar{L}Y), P(L\bar{Y}), \text{ and } P(\bar{L}\bar{Y})$$



$$P(L|Y)P(Y), P(\bar{L}|Y)P(Y), P(L|\bar{Y})P(\bar{Y}), \text{ and } P(\bar{L}|\bar{Y})P(\bar{Y})$$

# Transition probabilities

$$\begin{aligned} P(i \rightarrow j) &= P(i \rightarrow j | \text{"Recognize"})P(\text{"Recognize"}) \\ &+ P(i \rightarrow j | \text{"Erase"})P(\text{"Erase"}) \\ &+ P(i \rightarrow j | \text{"Reject"})P(\text{"Reject"}) \\ &+ P(i \rightarrow j | \text{"No Feedback"})P(\text{"No Feedback"}). \end{aligned}$$

# Transition probabilities

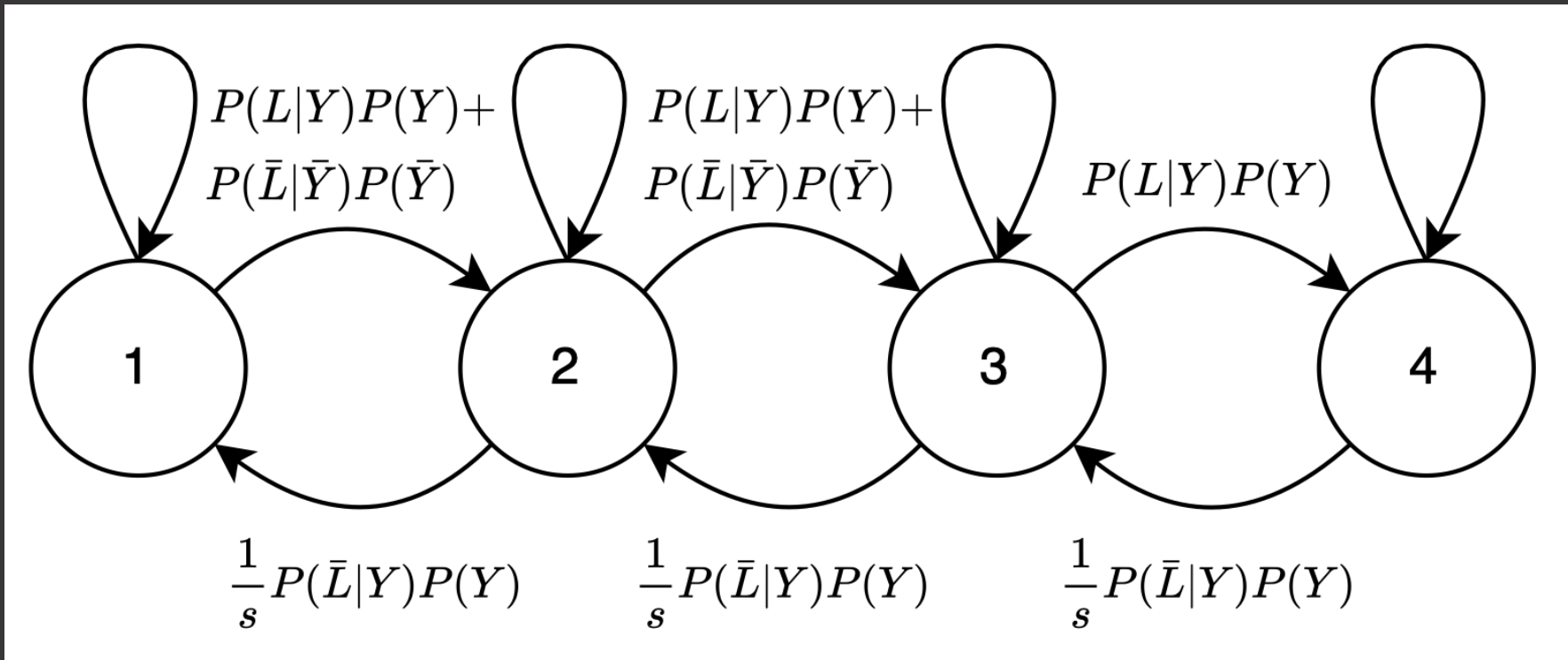
$$\begin{aligned}
 P(1 \rightarrow 2) &= P(1 \rightarrow 2|LY)P(L|Y)P(Y) \longleftarrow \text{Recognize} \\
 &+ P(1 \rightarrow 2|\bar{L}Y)P(\bar{L}|Y)P(Y) \longleftarrow \text{Erase} \\
 &+ P(1 \rightarrow 2|\bar{L}\bar{Y})P(\bar{L}|\bar{Y})P(\bar{Y}) \longleftarrow \text{Reject} \\
 &+ P(1 \rightarrow 2|L\bar{Y})P(L|\bar{Y})P(\bar{Y}).
 \end{aligned}$$

# Transition probabilities

$$\begin{aligned}
 P(1 \rightarrow 2) &= P(1 \rightarrow 2|LY)P(L|Y)P(Y) \longleftarrow \text{Recognize} \\
 &+ P(1 \rightarrow 2|\bar{L}Y)P(\bar{L}|Y)P(Y) \longleftarrow \text{Erase} \\
 &+ P(1 \rightarrow 2|\bar{L}\bar{Y})P(\bar{L}|\bar{Y})P(\bar{Y}) \longleftarrow \text{Reject} \\
 &+ P(1 \rightarrow 2|L\bar{Y})P(L|\bar{Y})P(\bar{Y}).
 \end{aligned}$$

$$\begin{aligned}
 P(1 \rightarrow 2) &= 1 \cdot P(L|Y)P(Y) \\
 &+ 0 \cdot P(\bar{L}|Y)P(Y) \\
 &+ 1 \cdot P(\bar{L}|\bar{Y})P(\bar{Y}) \\
 &+ 0 \cdot P(L|\bar{Y})P(\bar{Y}).
 \end{aligned}$$

# Complete transition graph

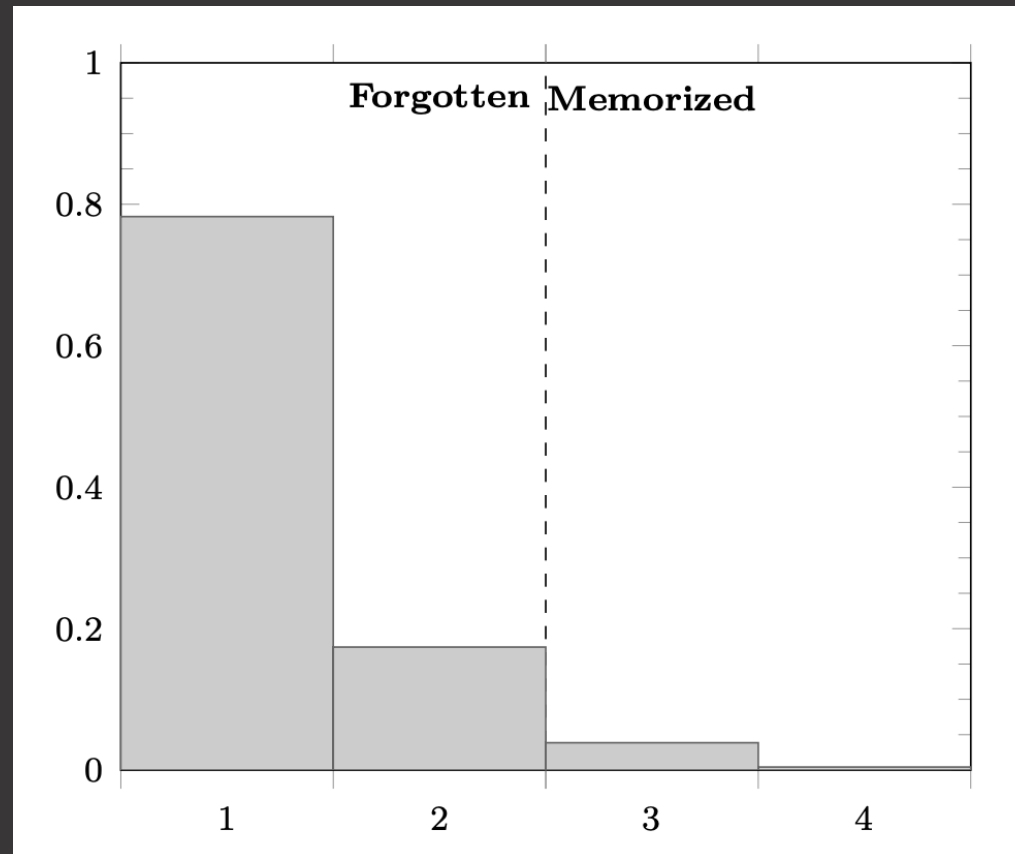


# Stationary distribution

|         |   | 1.       | 2.       | 3.               | 4.    | 5.                                              | 6.       |
|---------|---|----------|----------|------------------|-------|-------------------------------------------------|----------|
| $\pi_1$ | = | $\alpha$ |          | $P(\bar{L} Y)^3$ |       |                                                 | $P(Y)^2$ |
| $\pi_2$ | = | $\alpha$ |          | $P(\bar{L} Y)^2$ | $s$   | $(P(L Y)P(Y) + P(\bar{L} \bar{Y})P(\bar{Y}))$   | $P(Y)$   |
| $\pi_3$ | = | $\alpha$ |          | $P(\bar{L} Y)$   | $s^2$ | $(P(L Y)P(Y) + P(\bar{L} \bar{Y})P(\bar{Y}))^2$ |          |
| $\pi_4$ | = | $\alpha$ | $P(L Y)$ |                  | $s^3$ | $(P(L Y)P(Y) + P(\bar{L} \bar{Y})P(\bar{Y}))^2$ |          |

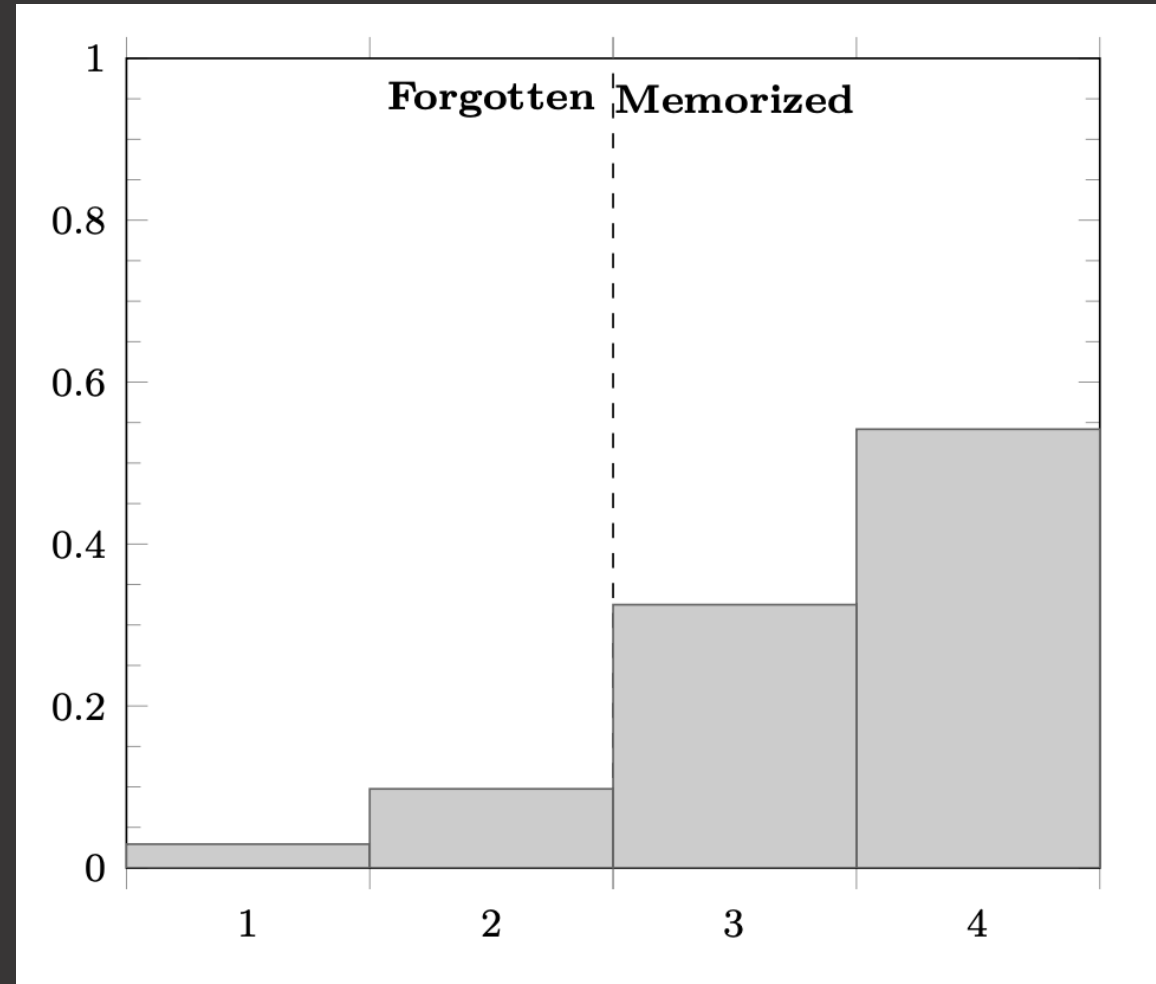
# Tsetlin machines forget infrequent patterns

*Forget Value: 1*  
 $P(Y|L) = 0.5$   
 $P(L) = 0.1$



# Forget value controls forgetting

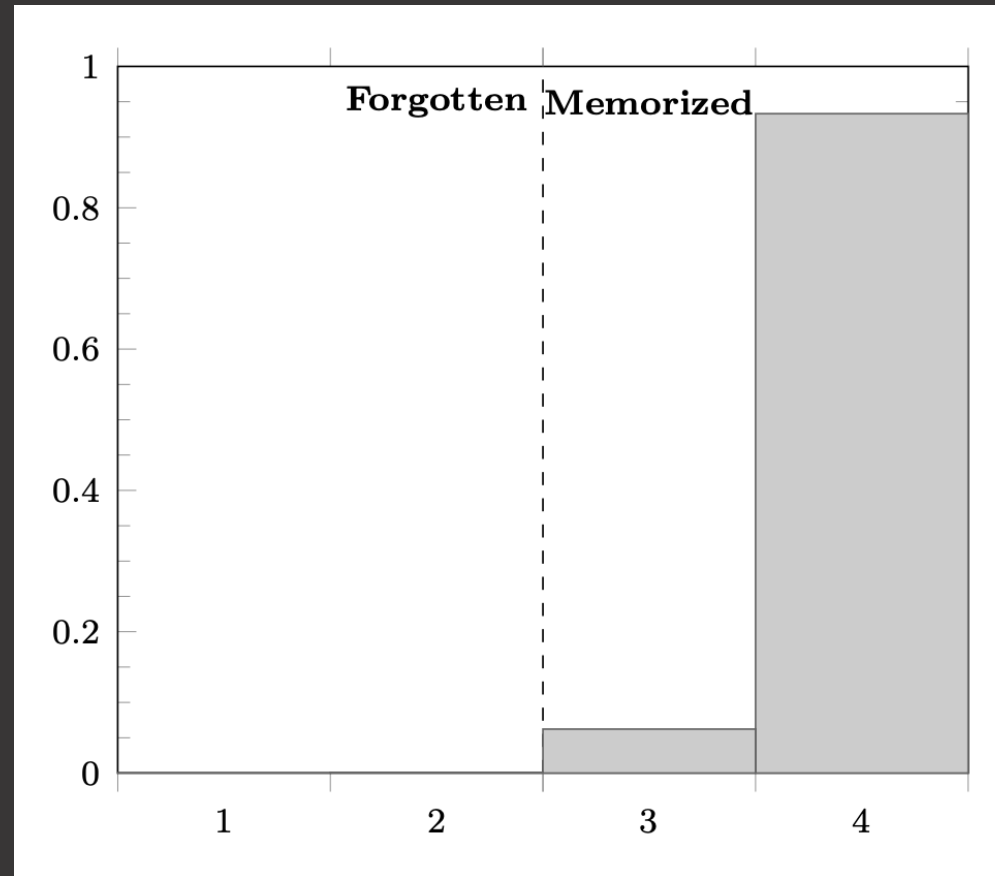
*Forget Value: 1/15*  
 $P(Y|L) = 0.5$





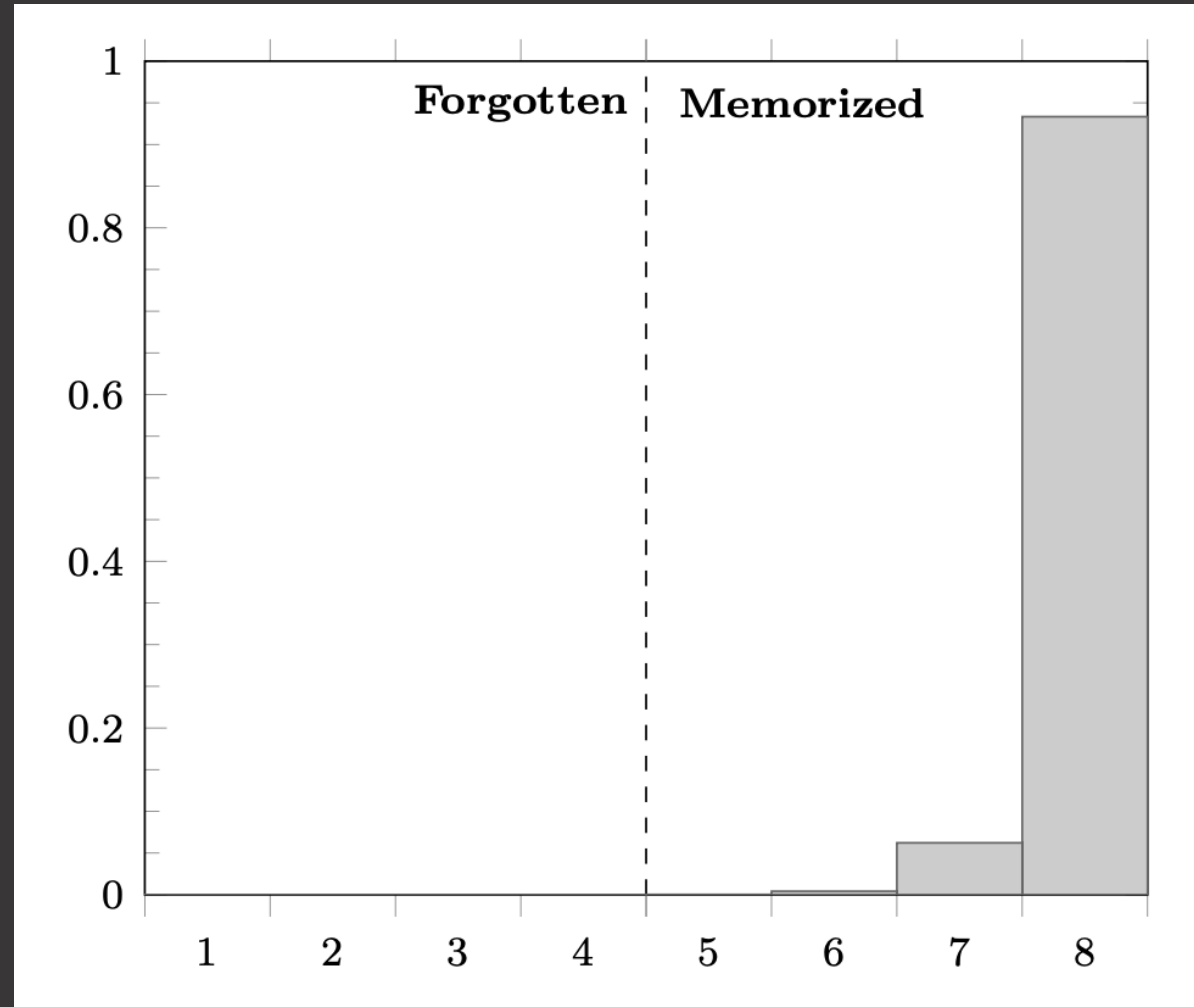
# Prioritizes discriminatory features

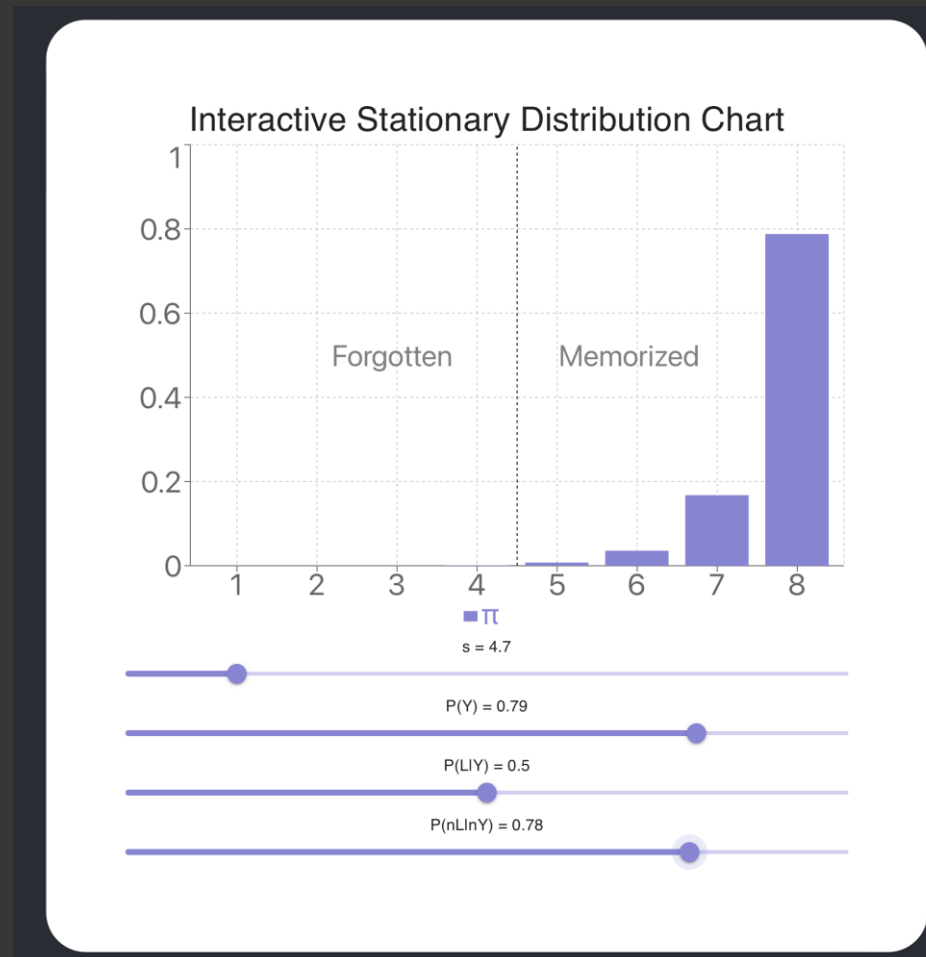
*Forget Value: 1/15*  
 $P(Y|L) = 0.9$



# Deeper memory increases learning accuracy

*Forget Value: 1/15*  
 $P(Y|L) = 0.9$





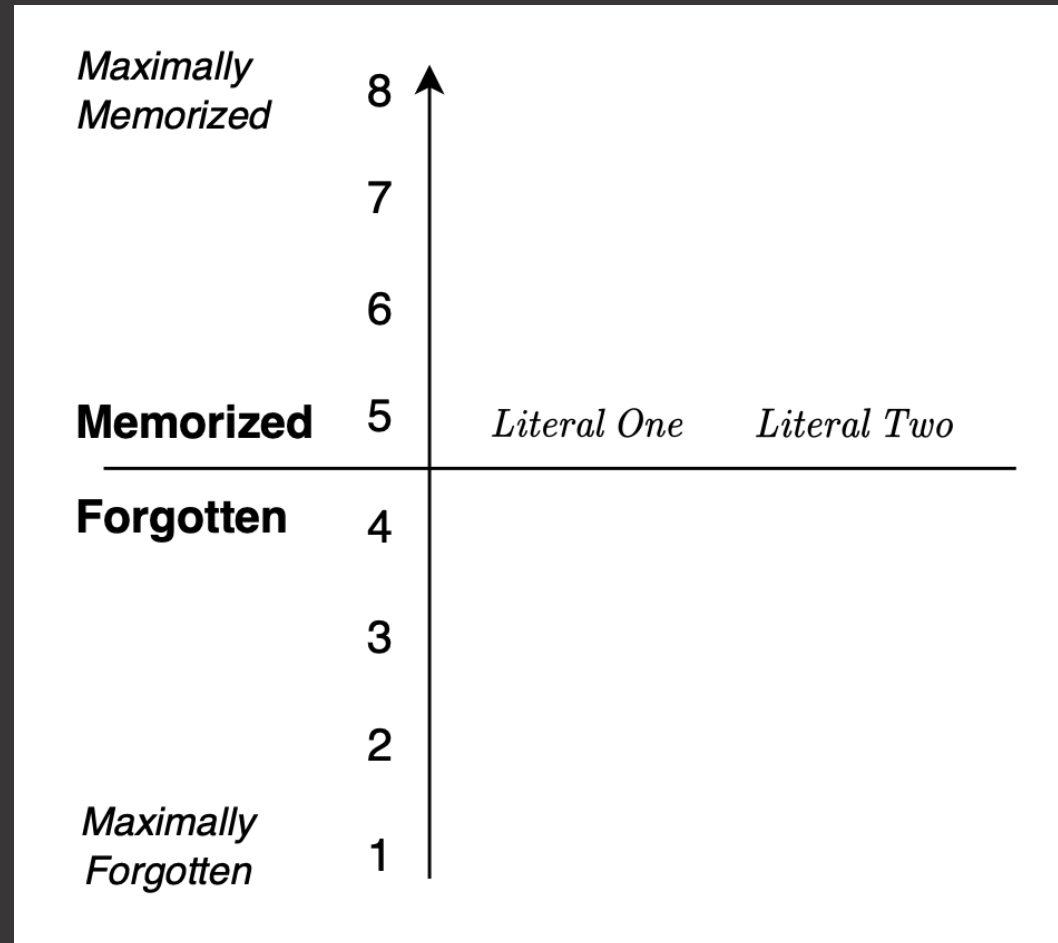
<https://isoporhode.github.io/interactive-stationary-distribution-chart/>

# Multi-literal learning

# Prisoner's dilemma

|              |               | Prisoner Two                           |                                        |
|--------------|---------------|----------------------------------------|----------------------------------------|
|              |               | Remain Silent                          | Betray                                 |
| Prisoner One | Remain Silent | <i>One Year</i><br><i>One Year</i>     | <i>Released</i><br><i>Twenty Years</i> |
|              | Betray        | <i>Twenty Years</i><br><i>Released</i> | <i>Five Years</i><br><i>Five Years</i> |

# Literal game



# Literal game

|             |           | Literal Two                 |                                        |             |           | Literal Two                        |                                    |
|-------------|-----------|-----------------------------|----------------------------------------|-------------|-----------|------------------------------------|------------------------------------|
|             |           | Forgotten                   | Memorized                              |             |           | Forgotten                          | Memorized                          |
| Literal One | Forgotten | $\emptyset$<br><i>True</i>  | $L_2$<br><i>True</i>                   | Literal One | Forgotten | $\uparrow\uparrow$<br>$\downarrow$ | $\uparrow\uparrow$<br>$\downarrow$ |
|             | Memorized | $\bar{L}_1$<br><i>False</i> | $\bar{L}_1 \wedge L_2$<br><i>False</i> |             | Memorized | $\downarrow$<br>$\downarrow$       | $\downarrow$<br>$\downarrow$       |

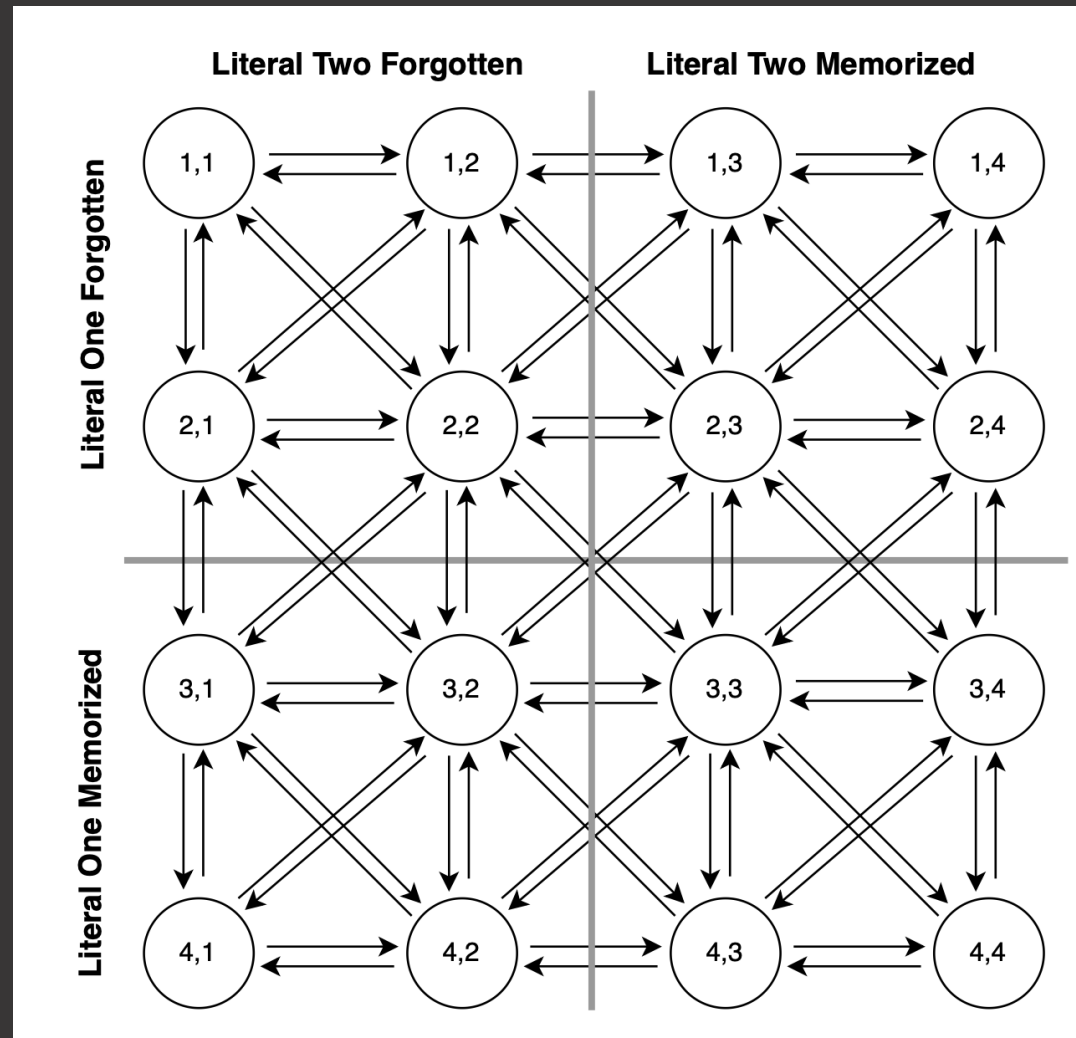
Observation  $(\bar{L}_1, L_2, Y)$

# Eight literal games

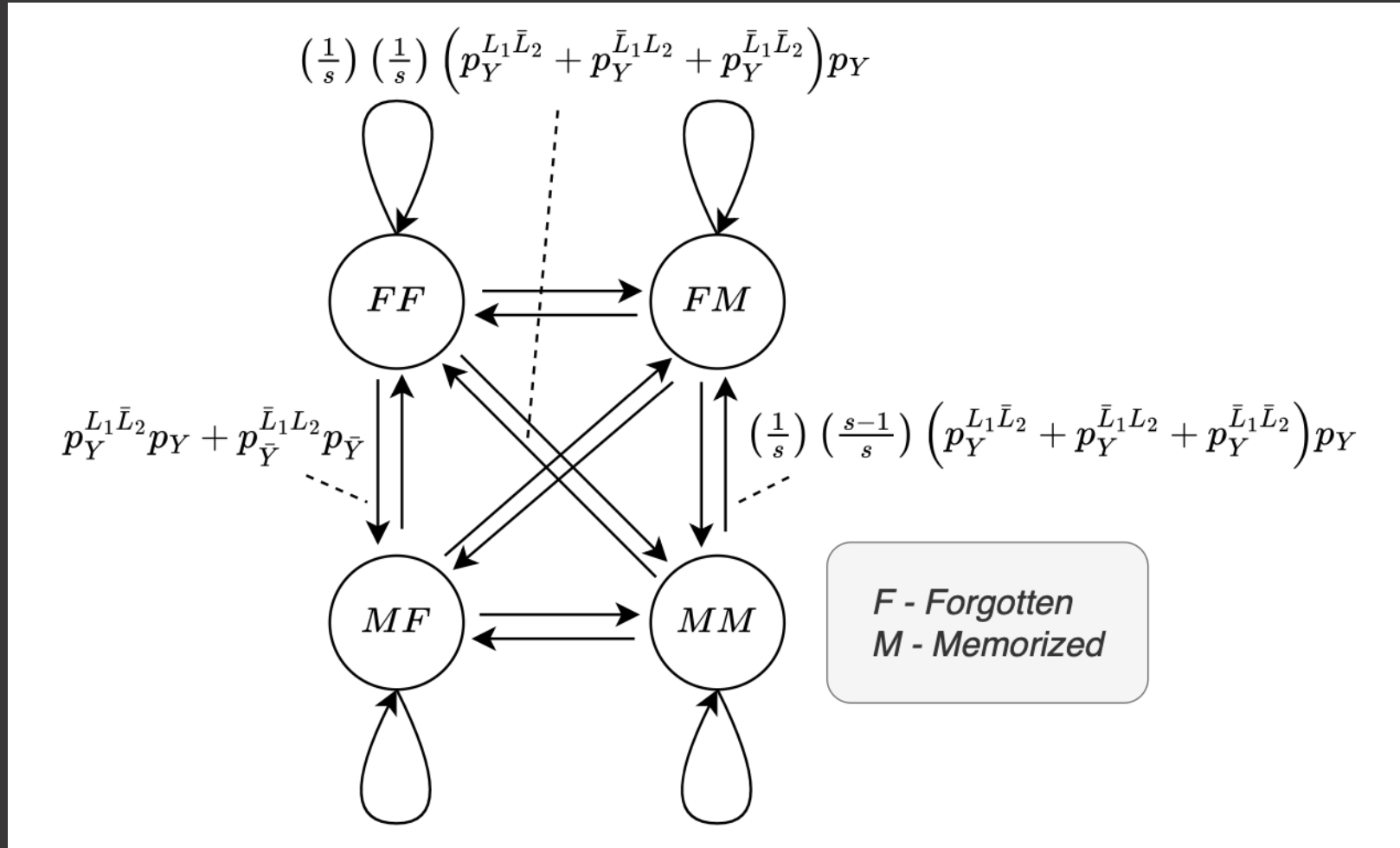
|                       |                       | Literal Two Forgotten             |                            |                        | Literal Two Memorized             |                                             |                        |
|-----------------------|-----------------------|-----------------------------------|----------------------------|------------------------|-----------------------------------|---------------------------------------------|------------------------|
| Literal One Forgotten | Literal One Forgotten | Obs.                              | Rule                       | Outcome                | Obs.                              | Rule                                        | Outcome                |
|                       |                       | $(L_1, L_2, Y)$                   | $\emptyset = \text{True}$  | $\uparrow\uparrow$     | $(L_1, L_2, Y)$                   | $L_2 = \text{True}$                         | $\uparrow\uparrow$     |
|                       |                       | $(L_1, \bar{L}_2, Y)$             | $\emptyset = \text{True}$  | $\uparrow\downarrow$   | $(L_1, \bar{L}_2, Y)$             | $\bar{L}_2 = \text{False}$                  | $\downarrow\downarrow$ |
|                       |                       | $(\bar{L}_1, L_2, Y)$             | $\emptyset = \text{True}$  | $\downarrow\uparrow$   | $(\bar{L}_1, L_2, Y)$             | $L_2 = \text{True}$                         | $\downarrow\uparrow$   |
|                       |                       | $(\bar{L}_1, \bar{L}_2, Y)$       | $\emptyset = \text{True}$  | $\downarrow\downarrow$ | $(\bar{L}_1, \bar{L}_2, Y)$       | $\bar{L}_2 = \text{False}$                  | $\downarrow\downarrow$ |
|                       |                       | $(L_1, L_2, \bar{Y})$             | $\emptyset = \text{True}$  | $--$                   | $(L_1, L_2, \bar{Y})$             | $L_2 = \text{True}$                         | $--$                   |
|                       |                       | $(L_1, \bar{L}_2, \bar{Y})$       | $\emptyset = \text{True}$  | $- \uparrow$           | $(L_1, \bar{L}_2, \bar{Y})$       | $\bar{L}_2 = \text{False}$                  | $--$                   |
|                       |                       | $(\bar{L}_1, L_2, \bar{Y})$       | $\emptyset = \text{True}$  | $\uparrow -$           | $(\bar{L}_1, L_2, \bar{Y})$       | $L_2 = \text{True}$                         | $\uparrow -$           |
| Literal One Memorized | Literal One Memorized | $(\bar{L}_1, \bar{L}_2, \bar{Y})$ | $\emptyset = \text{True}$  | $\uparrow\uparrow$     | $(\bar{L}_1, \bar{L}_2, \bar{Y})$ | $\bar{L}_2 = \text{False}$                  | $--$                   |
|                       |                       | Obs.                              | Rule                       | Outcome                | Obs.                              | Rule                                        | Outcome                |
|                       |                       | $(L_1, L_2, Y)$                   | $L_1 = \text{True}$        | $\uparrow\uparrow$     | $(L_1, L_2, Y)$                   | $L_1 \wedge L_2 = \text{True}$              | $\uparrow\uparrow$     |
|                       |                       | $(L_1, \bar{L}_2, Y)$             | $L_1 = \text{True}$        | $\uparrow\downarrow$   | $(L_1, \bar{L}_2, Y)$             | $L_1 \wedge \bar{L}_2 = \text{False}$       | $\downarrow\downarrow$ |
|                       |                       | $(\bar{L}_1, L_2, Y)$             | $\bar{L}_1 = \text{False}$ | $\downarrow\downarrow$ | $(\bar{L}_1, L_2, Y)$             | $\bar{L}_1 \wedge L_2 = \text{False}$       | $\downarrow\downarrow$ |
|                       |                       | $(\bar{L}_1, \bar{L}_2, Y)$       | $\bar{L}_1 = \text{False}$ | $\downarrow\downarrow$ | $(\bar{L}_1, \bar{L}_2, Y)$       | $\bar{L}_1 \wedge \bar{L}_2 = \text{False}$ | $\downarrow\downarrow$ |
|                       |                       | $(L_1, L_2, \bar{Y})$             | $L_1 = \text{True}$        | $--$                   | $(L_1, L_2, \bar{Y})$             | $L_1 \wedge L_2 = \text{True}$              | $--$                   |
|                       |                       | $(L_1, \bar{L}_2, \bar{Y})$       | $L_1 = \text{True}$        | $- \uparrow$           | $(L_1, \bar{L}_2, \bar{Y})$       | $L_1 \wedge \bar{L}_2 = \text{False}$       | $--$                   |
|                       |                       | $(\bar{L}_1, L_2, \bar{Y})$       | $\bar{L}_1 = \text{False}$ | $--$                   | $(\bar{L}_1, L_2, \bar{Y})$       | $\bar{L}_1 \wedge L_2 = \text{False}$       | $--$                   |
|                       |                       | $(\bar{L}_1, \bar{L}_2, \bar{Y})$ | $\bar{L}_1 = \text{False}$ | $--$                   | $(\bar{L}_1, \bar{L}_2, \bar{Y})$ | $\bar{L}_1 \wedge \bar{L}_2 = \text{False}$ | $--$                   |



# Transition graph of literal game



# Transition graph of literal game



# Tsetlin machines select informative literals

*Forget Value: 1*

$$P(Y|L_1) = 0.5$$

$$P(Y|L_2) = 0.9$$

|                |   | Literal Two Forgotten |     |     |     | Literal Two Memorized |     |     |     |
|----------------|---|-----------------------|-----|-----|-----|-----------------------|-----|-----|-----|
|                |   | 1                     | 2   | 3   | 4   | 5                     | 6   | 7   | 8   |
| Lit. One Forg. | 1 | .00                   | .00 | .00 | .01 | .02                   | .06 | .11 | .16 |
|                | 2 | .00                   | .00 | .00 | .00 | .01                   | .02 | .05 | .14 |
|                | 3 | .00                   | .00 | .00 | .00 | .01                   | .01 | .03 | .10 |
|                | 4 | .00                   | .00 | .00 | .01 | .01                   | .01 | .02 | .05 |
| Lit. One Mem.  | 5 | .00                   | .00 | .00 | .00 | .01                   | .01 | .01 | .03 |
|                | 6 | .00                   | .00 | .00 | .00 | .00                   | .01 | .01 | .01 |
|                | 7 | .00                   | .00 | .00 | .00 | .00                   | .00 | .01 | .01 |
|                | 8 | .00                   | .00 | .00 | .00 | .00                   | .00 | .00 | .01 |

# Tsetlin machines learn multiple alternatives

*Forget Value: 1*  
 $P(Y|L_1) = 0.9$   
 $P(Y|L_2) = 0.9$

|                |   | Literal Two Forgotten |     |     |     | Literal Two Memorized |     |     |     |
|----------------|---|-----------------------|-----|-----|-----|-----------------------|-----|-----|-----|
|                |   | 1                     | 2   | 3   | 4   | 5                     | 6   | 7   | 8   |
| Lit. One Forg. | 1 | .00                   | .00 | .00 | .01 | .02                   | .04 | .06 | .07 |
|                | 2 | .00                   | .00 | .00 | .00 | .01                   | .02 | .03 | .06 |
|                | 3 | .00                   | .00 | .00 | .00 | .01                   | .01 | .02 | .04 |
|                | 4 | .01                   | .00 | .00 | .01 | .01                   | .01 | .01 | .02 |
| Lit. One Mem.  | 5 | .02                   | .01 | .01 | .01 | .01                   | .00 | .01 | .01 |
|                | 6 | .04                   | .02 | .01 | .01 | .00                   | .01 | .00 | .01 |
|                | 7 | .06                   | .03 | .02 | .01 | .01                   | .00 | .01 | .00 |
|                | 8 | .07                   | .06 | .04 | .02 | .01                   | .01 | .00 | .01 |

# Tsetlin machines pick the best literal

*Forget Value: 1*  
 $P(Y|L_1) = 0.7$   
 $P(Y|L_2) = 0.9$

|                |   | Literal Two Forgotten |     |     |     | Literal Two Memorized |     |     |     |
|----------------|---|-----------------------|-----|-----|-----|-----------------------|-----|-----|-----|
|                |   | 1                     | 2   | 3   | 4   | 5                     | 6   | 7   | 8   |
| Lit. One Forg. | 1 | .00                   | .00 | .00 | .01 | .02                   | .05 | .09 | .12 |
|                | 2 | .00                   | .00 | .00 | .00 | .01                   | .02 | .05 | .11 |
|                | 3 | .00                   | .00 | .00 | .00 | .01                   | .01 | .02 | .08 |
|                | 4 | .00                   | .00 | .00 | .01 | .01                   | .01 | .02 | .04 |
| Lit. One Mem.  | 5 | .01                   | .01 | .01 | .01 | .02                   | .01 | .01 | .02 |
|                | 6 | .01                   | .01 | .01 | .01 | .00                   | .01 | .01 | .01 |
|                | 7 | .01                   | .01 | .01 | .01 | .01                   | .00 | .01 | .01 |
|                | 8 | .01                   | .01 | .01 | .01 | .01                   | .00 | .00 | .01 |

# Tsetlin machines pick the best literal

*Forget Value: 1*  
 $P(Y|L_1) = 0.7$   
 $P(Y|L_2) = 0.9$

|                |   | Literal Two Forgotten |     |     |     | Literal Two Memorized |     |     |     |
|----------------|---|-----------------------|-----|-----|-----|-----------------------|-----|-----|-----|
|                |   | 1                     | 2   | 3   | 4   | 5                     | 6   | 7   | 8   |
| Lit. One Forg. | 1 | .00                   | .00 | .00 | .01 | .02                   | .05 | .09 | .12 |
|                | 2 | .00                   | .00 | .00 | .00 | .01                   | .02 | .05 | .11 |
|                | 3 | .00                   | .00 | .00 | .00 | .01                   | .01 | .02 | .08 |
|                | 4 | .00                   | .00 | .00 | .01 | .01                   | .01 | .02 | .04 |
| Lit. One Mem.  | 5 | .01                   | .01 | .01 | .01 | .02                   | .01 | .01 | .02 |
|                | 6 | .01                   | .01 | .01 | .01 | .00                   | .01 | .01 | .01 |
|                | 7 | .01                   | .01 | .01 | .01 | .01                   | .00 | .01 | .01 |
|                | 8 | .01                   | .01 | .01 | .01 | .01                   | .00 | .00 | .01 |

# Higher forget value creates longer rules

*Forget Value: 1/4*

$$P(Y|L_1) = 0.9$$

$$P(Y|L_2) = 0.9$$

|                |   | Literal Two Forgotten |     |     |     | Literal Two Memorized |     |     |     |
|----------------|---|-----------------------|-----|-----|-----|-----------------------|-----|-----|-----|
|                |   | 1                     | 2   | 3   | 4   | 5                     | 6   | 7   | 8   |
| Lit. One Forg. | 1 | .00                   | .00 | .00 | .00 | .00                   | .00 | .00 | .00 |
|                | 2 | .00                   | .00 | .00 | .00 | .00                   | .00 | .00 | .00 |
|                | 3 | .00                   | .00 | .00 | .00 | .00                   | .00 | .00 | .01 |
|                | 4 | .00                   | .00 | .00 | .00 | .00                   | .00 | .01 | .01 |
| Lit. One Mem.  | 5 | .00                   | .00 | .00 | .00 | .01                   | .01 | .02 | .02 |
|                | 6 | .00                   | .00 | .00 | .00 | .01                   | .02 | .04 | .04 |
|                | 7 | .00                   | .00 | .00 | .01 | .02                   | .04 | .08 | .11 |
|                | 8 | .00                   | .00 | .01 | .01 | .02                   | .04 | .11 | .36 |

# Multi-rule learning



# Two-rule game with two literals

|             |              | Rule Two             |                      |                                      |                                      |                      |
|-------------|--------------|----------------------|----------------------|--------------------------------------|--------------------------------------|----------------------|
|             |              | $\mathbf{x_2}$       | <i>Forgotten</i>     |                                      | <i>Memorized</i>                     |                      |
|             |              | $\mathbf{x_1}$       | <i>Forgotten</i>     | <i>Memorized</i>                     | <i>Forgotten</i>                     | <i>Memorized</i>     |
| Rule One    | <i>Forg.</i> | <i>Forg.</i>         | R1: $\emptyset$      | R1: $\emptyset$                      | R1: $\emptyset$                      | R1: $\emptyset$      |
|             |              |                      | R2: $\emptyset$      | R2: $x_1$                            | R2: $x_2$                            | R2: $x_1 \wedge x_2$ |
|             | <i>Mem.</i>  | <i>Mem.</i>          | R1: $x_1$            | R1: $x_1$                            | <b>R1: <math>\mathbf{x_1}</math></b> | R1: $x_1$            |
|             |              |                      | R2: $\emptyset$      | R2: $x_1$                            | <b>R2: <math>\mathbf{x_2}</math></b> | R2: $x_1 \wedge x_2$ |
|             | <i>Mem.</i>  | <i>Forg.</i>         | R1: $x_2$            | <b>R1: <math>\mathbf{x_2}</math></b> | R1: $x_2$                            | R1: $x_2$            |
|             |              |                      | R2: $\emptyset$      | <b>R2: <math>\mathbf{x_1}</math></b> | R2: $x_2$                            | R2: $x_1 \wedge x_2$ |
| <i>Mem.</i> | <i>Mem.</i>  | R1: $x_1 \wedge x_2$ | R1: $x_1 \wedge x_2$ | R1: $x_1 \wedge x_2$                 | R1: $x_1 \wedge x_2$                 |                      |
|             |              | R2: $\emptyset$      | R2: $x_1$            | R2: $x_2$                            | R2: $x_1 \wedge x_2$                 |                      |

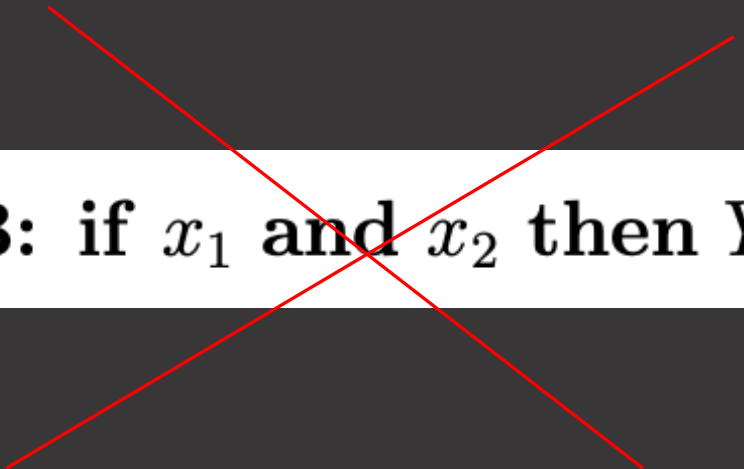
# Example learning outcome

$$Y = x_1 \text{ or } x_2.$$

**R1:** if  $x_1$  then  $Y$ .

**R2:** if  $x_2$  then  $Y$ .

**R3:** if  $x_1$  and  $x_2$  then  $Y$ .



|          |                | Rule Two       |   |   |   |   |   |   |     |   |   |     |     |     |     |     |     |   |
|----------|----------------|----------------|---|---|---|---|---|---|-----|---|---|-----|-----|-----|-----|-----|-----|---|
| Rule One | x <sub>2</sub> | 1              |   |   |   | 2 |   |   |     | 3 |   |     |     | 4   |     |     |     |   |
|          |                | x <sub>1</sub> | 1 | 2 | 3 | 4 | 1 | 2 | 3   | 4 | 1 | 2   | 3   | 4   | 4   | 3   | 2   | 1 |
|          |                |                |   |   |   |   |   |   |     |   |   |     |     |     |     |     |     |   |
|          | 1              | 1              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |
|          |                | 2              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |
|          |                | 3              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |
|          |                | 4              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |
|          | 2              | 1              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |
|          |                | 2              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |
|          |                | 3              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |
|          |                | 4              | 0 | 0 | 0 | 0 | 0 | 0 | .01 | 0 | 0 | .01 | .01 | .01 | .01 | .01 | .01 | 0 |
|          | 3              | 1              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |
|          |                | 2              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |
|          |                | 3              | 0 | 0 | 0 | 0 | 0 | 0 | .01 | 0 | 0 | .01 | .01 | .02 | .01 | .01 | .01 | 0 |
|          |                | 4              | 0 | 0 | 0 | 0 | 0 | 0 | .01 | 0 | 0 | .01 | .02 | .05 | .02 | .01 | .01 | 0 |
|          | 4              | 4              | 0 | 0 | 0 | 0 | 0 | 0 | .01 | 0 | 0 | .02 | .05 | .31 | .05 | .01 | .01 | 0 |
|          |                | 3              | 0 | 0 | 0 | 0 | 0 | 0 | .01 | 0 | 0 | .01 | .02 | .05 | .02 | .01 | .01 | 0 |
|          |                | 2              | 0 | 0 | 0 | 0 | 0 | 0 | .01 | 0 | 0 | .01 | .01 | .01 | .01 | .01 | .01 | 0 |
|          |                | 1              | 0 | 0 | 0 | 0 | 0 | 0 | 0   | 0 | 0 | 0   | 0   | 0   | 0   | 0   | 0   | 0 |

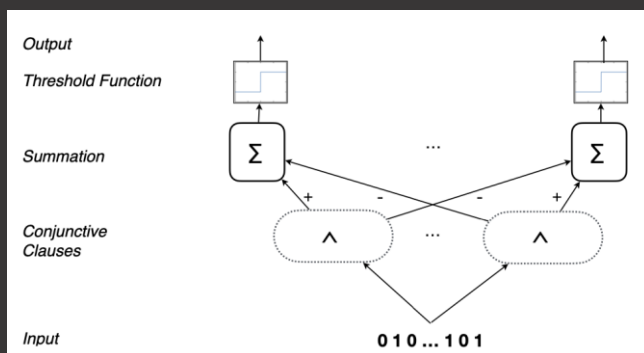
Without voting margin

## Rule One

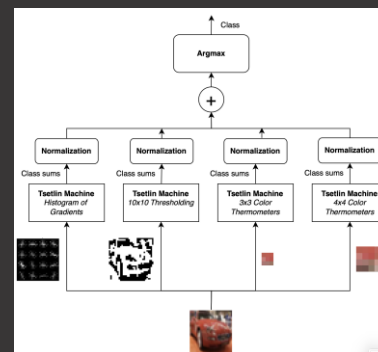
|                |                | Rule Two |   |     |     |     |     |   |   |   |   |     |     |     |     |   |   |
|----------------|----------------|----------|---|-----|-----|-----|-----|---|---|---|---|-----|-----|-----|-----|---|---|
| x <sub>2</sub> |                | 1        |   |     |     | 2   |     |   |   | 3 |   |     |     | 4   |     |   |   |
|                | x <sub>1</sub> | 1        | 2 | 3   | 4   | 4   | 3   | 2 | 1 | 4 | 3 | 2   | 1   | 1   | 2   | 3 | 4 |
| 1              | 1              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
|                | 2              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
|                | 3              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | .01 | .04 | .01 | 0 | 0 |
|                | 4              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | .01 | .04 | .16 | .04 | 0 | 0 |
| 2              | 4              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | .01 | .04 | .01 | 0 | 0 |
|                | 3              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | .01 | 0   | 0 | 0 |
|                | 2              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
|                | 1              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
| 3              | 4              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
|                | 3              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
|                | 2              | 0        | 0 | 0   | .01 | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
|                | 1              | 0        | 0 | .01 | .04 | .01 | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
| 4              | 1              | 0        | 0 | .04 | .16 | .04 | .01 | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
|                | 2              | 0        | 0 | .01 | .04 | .01 | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
|                | 3              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |
|                | 4              | 0        | 0 | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0   | 0   | 0   | 0   | 0 | 0 |

# Next lecture:

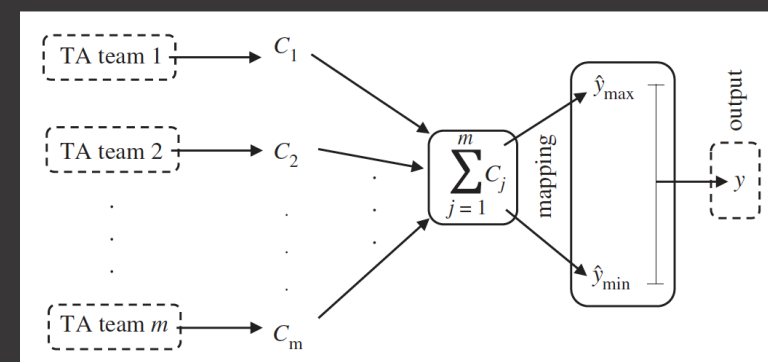
## Multi-output with rule sharing



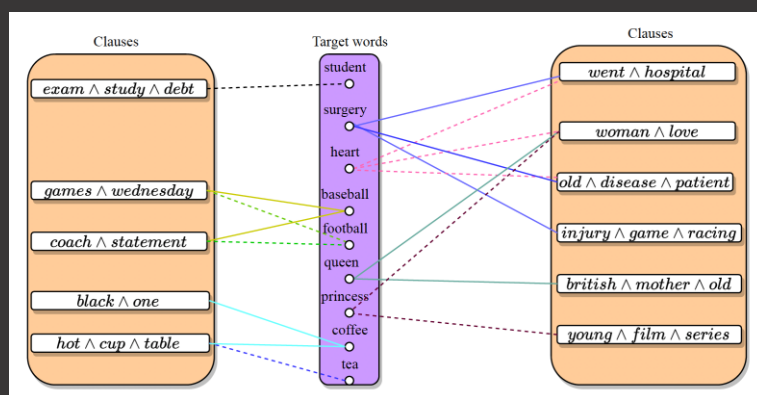
## Composites



## Regression



## Word embedding



## Hypervectors

